

stack

```
#include <iostream>

#include <string>

using namespace std;

class stack{

private:

int top;

int arr[5];

public:

stack ()//constructor

{

top = -1;

for(int i=0;i<5;i++)

arr[i]=0;

}

bool isEmpty()

{

if(top== -1)

return true;

else return false;

}

bool isFull()

{

if(top== 4)

return true;

else return false;

}

void push(int val)

{
```

```
if(isFull())
{
cout<<"stack overflow"<<endl;
}
else
{
top++;
arr[top]= val;

}
}
int pop()
{
if(isEmpty())
{
cout<<"stack underflow"<<endl;
}
else
{
int popval = arr[top];
arr[top] = 0;
top--;
return popval;
}
}
int count()
{
return (top + 1);
}
int peak(int pos)
{
```

```

if(isEmpty())
{
cout<<"stack underflow"<<endl;
}
else
{
return arr[pos];

}
}

void change(int pos, int val)
{
arr[pos] = val;
cout<<"Value changed at location"<<pos<<endl;
}

void display()
{
cout<<"All values in the stack are"<<endl;
for(int i = 4;i>=0; i--)
{
cout<<arr[i]<<endl;
}
}

};

int main()
{
stack s1;//object
int option, position,value;
do
{

```

```

cout<<"what operations do you want to perform? select option number.Enter 0 to exit"<<endl;
cout<<"1. push()"<<endl;
cout<<"2. pop()"<<endl;
cout<<"3. isEmpty()"<<endl;
cout<<"4. isFull()"<<endl;
cout<<"5. peak()"<<endl;
cout<<"6. count()"<<endl;
cout<<"7. change()"<<endl;
cout<<"8. display()"<<endl;
cout<<"9. clear screen"<<endl<<endl;

cin>>option;
switch(option)
{
case 1:
cout<<"Enter an item to push in the stack"<<endl;
cin>>value;
s1.push(value);
break;
case 2:
cout<<"Pop Function called - pop value:"<<s1.pop()<<endl;
break;
case 3:
if(s1.isEmpty())
cout<<"Stack is empty"<<endl;
break;
case 4:
if(s1.isFull())
cout<<"Stack is Full"<<endl;
else
cout<<"Stack is not Full"<<endl;
break;

```

case 5:

```
cout<<"Enter position of item you want to peak"<<endl;
```

```
cin>>position;
```

```
cout<<"Peak function called - value at position"<<position<<"is"<<s1.peak(position)<<endl;
```

```
break;
```

case 6:

```
cout<<"count function called "<<"number of items in the stack are"<<s1.count()<<endl;
```

```
break;
```

case 7:

```
cout<<"change function called"<<endl;
```

```
cout<<"Enter position of item you want to change";
```

```
cin>>position;
```

```
cout<<endl;
```

```
cout<<"Enter value of Item you want to change";
```

```
cin>>value;
```

```
s1.change(position, value);
```

```
break;
```

case 8:

```
cout<<"Display function is called"<<endl;
```

```
s1.display();
```

```
break;
```

case 9:

```
system("cls");
```

```
break;
```

default:

```
cout<<"Enter proper option number"<<endl;
```

```
}
```

```
}
```

```
while(option!=0);
```

```
return 0;
```

}